Paper 1: Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. https://doi.org/10.1016/j.ijmedinf.2025.105838

Pass 1

Problem: There is limited understanding and evidence regarding the impact of artificial intelligence (AI) and machine learning (ML) on emergency department (ED) triage processes.

Method: This scoping review was conducted following the Joanna Briggs Institute (JBI) methodology for scoping reviews.

Outcome: Machine learning models outperformed traditional triage systems by improving predictive accuracy, disease identification, and risk assessment. They effectively determined hospitalization needs, optimized resource allocation, and enhanced patient care, including predicting length of stay. Overall, ML shows strong potential to redefine triage precision in emergency departments.

Pass 2

Limitation: The study's exclusion criteria relied on subjective judgments when screening titles and abstracts, which may have introduced inconsistencies. Additionally, excluding studies with higher bias scores during appraisal could have led to the loss of potentially valuable insights.

Paper 2: TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers
https://doi.org/10.1016/j.procs.2024.11.130

Pass 1

Problem:
ED overcrowding combined with aging population and case variability in human triage. The authors want to leverage AI/ML to enhance the triaging process.

Method:
Seven ML models (SVM, RF, ANN, GBM, LR, XGBoost, plus a stacking ensemble) were trained to predict patient priority using the Korean Triage and Acuity Scale (KTAS).

Outcome:
Stacking model: 80.05% accuracy. Best individuals: GBM 79%, SVM 78.7%. AUC ≈ 0.93 for the stacker.

Pass 2

Limitation:
A key limitation is the reliance on a single dataset, which may hinder the generalizability of the models to other clinical environments with varying demographics, seasonal factors, and hospital contexts.

Paper 3: AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. https://doi.org/10.1016/j.ijmedinf.2025.105838

Pass 1

Problem: Traditional triage systems, while structured, rely on subjective assessments, which can lack consistency during peak hours or mass casualty events.

Method:
Analyzed peer-reviewed articles published between 2015 and 2024, identified through searches in PubMed, Scopus, IEEE Xplore, and Google Scholar. Findings were synthesized to provide a comprehensive overview of their potential and limitations.

Outcome: The review identifies substantial benefits of AI-driven triage, including improved patient prioritization, reduced wait times, and optimized resource allocation. However, challenges such as data quality issues, algorithmic bias, clinician trust, and ethical concerns are significant barriers to widespread adoption.

Pass 2

Limitation: The review is narrative in nature, which inherently lacks the methodological rigour of systematic reviews or meta-analyses. As a result, while it provides a broad synthesis of the existing literature, it does not quantitatively evaluate the strength of evidence or statistical relationships.

Paper 4: Evolution of the "Internet Plus Health Care" Mode Enabled by Artificial Intelligence: Development and Application of an Outpatient Triage System
https://www.jmir.org/2024/1/e51711/

Pass 1

Problem: Although new technologies have increased the efficiency and convenience of medical care, patients still struggle to identify specialized outpatient departments in Chinese tertiary hospitals due to a lack of medical knowledge.

Method: Collected 395,790 electronic medical records (EMRs) and 500 medical dialogue groups. The EMRs were divided into 3 data sets to design and train the triage model (n=387,876, 98%) and test (n=3957, 1%) and validate (n=3957, 1%) it.

Outcome: The triage system isn't as accurate as human nurses when it comes to pinpointing the exact department, but it has strengths: it can suggest more specialized departments, works faster, and improves hospital efficiency by reducing cancellations. In practice, it complements nurses by handling subspecialty recommendations and speeding up the triage process, while nurses remain superior in overall accuracy.

Pass 2

Limitation: One of many limitations is the limited training data. EMRs were collected from a single center, Xinhua Hospital, in which the departments for pediatric diseases may be more specialized than those in other hospitals.

Gap in Three-Step Format

ATTEMPTED
AI-enabled outpatient triage system trained on 395,790 EMRs at Xinhua Hospital improved efficiency by suggesting subspecialty departments faster than nurses.

MISSED:
Training data limited to a single center with unusually specialized pediatric departments; generalisability to hospitals without subspecialty divisions and mixed patient streams remains unproven.

WHY MERCER
Mercer Hospital's Caribbean outpatient context differs significantly from Xinhua, fewer subspecialty divisions, different patient mix, and seasonal illness patterns. A pilot validation at Mercer directly addresses the limitation by testing whether the AI triage system adapts to a non-Chinese, resource-constrained hospital environment.

Paper 5: Improving triage performance in emergency departments using machine learning and natural language processing: a systematic review https://doi.org/10.1186/s12873-024-01135-2

Pass 1

Problem: Traditional triage systems, reliant on human judgment, are prone to under-triage and over-triage, resulting in variability, bias, and incorrect patient classification.

Method: Researchers followed Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines, they conducted a systematic review across five databases: Web of Science, PubMed, Scopus, IEEE Xplore, and ACM Digital Library, from their inception of each database to October 2023. The risk of bias was assessed using the Prediction model Risk of Bias Assessment Tool (PROBAST). Only articles employing at least one ML and/or NLP method for patient triage classification were included.

Outcome: Sixty studies covering 57 ML algorithms were included. Logistic Regression (LR) was the most used model, while eXtreme Gradient Boosting (XGBoost), decision tree-based algorithms with Gradient Boosting (GB), and Deep Neural Networks (DNNs) showed superior performance .Frequent predictive variables included demographics and vital signs, with oxygen saturation, chief complaints, systolic blood pressure, age, and mode of arrival being the most retained. The ML algorithms showed significant bias risk due to critical bias assessment in classification models.

Pass 2

Limitation: NLP methods improved ML algorithms' classification capability using triage nursing and medical notes and structured clinical data compared to algorithms using only structured data. Feature engineering (FE) and class imbalance correction methods enhanced ML workflows' performance, but FE and eXplainable Artificial Intelligence (XAI) were underexplored in this field

Gap in Three-Step Format

ATTEMPTED
Systematic review of 60 ML/NLP triage studies found XGBoost, GB, and DNNs outperform traditional models. NLP improved classification when combined with structured data.

MISSED
Robust exploration of explainability, workflow integration, and generalisability across diverse healthcare settings is lacking.
Despite accuracy gains, most triage studies did not explore feature engineering to handle incomplete or imbalanced data, nor explainable AI to make model decisions transparent to clinicians. As a result, workflows remain vulnerable to bias and lack clinician trust

WHY MERCER
Mercer ED relies on nurse-led ESI triage with re-triage checks and has structured intake records but incomplete deeper notes. Mercer ED's nurse-led triage, re-triage workflow, and unique Caribbean demographics provide a tractable context to test explainability and integration, directly addressing the limitation.

Problem Statement

Traditional triage systems face under triage and over-triage rates of 20–30%, contributing to ED crowding and delayed care. Machine learning models achieved up to 80% accuracy (Araouchi & Adda, 2024), yet most lack explainability and fail in resource constrained settings (Porto, 2024). At Mercer General, a Caribbean ED serving 80,000 residents with surges over 150 patients daily, nurse-led ESI triage remains paper-based, with no structured data or validated AI support. We propose an explainable, context fit ML triage tool trained on Mercer's digitised intake data and validated for nurse-led use.